

Srivatsan Sarvesan

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EDUCATION

University of California San Diego

M.S. in Machine Learning & Data Science (ECE)

La Jolla, CA, USA

Sept 2025 – Mar 2027

Courses: Optimization & Acceleration of DL on HW Platforms, Statistical Learning I, Trustworthy ML, Deep Generative Models

National Institute of Technology, Tiruchirappalli

Tiruchirappalli, India

B.Tech (Honors), Electronics & Communication Engg. - GPA: **3.96/4.0**, Minor in CSE

Jul 2021 – May 2025

Exchange Program, Dept. of Data Science & AI, IIT Madras

Jul 2024 – May 2025

Courses: Deep Learning, Pattern Recognition, Medical Image Analysis, Mapping Signal Processing Algos to DSP Arch, VLSI Systems

TECHNICAL SKILLS

Languages & Frameworks: Python, C++, CUDA, SQL, PyTorch, Hugging Face (Transformers, PEFT, TRL), TensorFlow, OpenCV

Post-Training & Agents: SFT, DPO, GRPO, LoRA/QLoRA, LangGraph, tool calling, RAG, LLM & agent evaluation

Inference & ML Sys: vLLM, FlashAttention, FlashInfer, torch.compile, DDP, quantization, distillation, torch.profiler, Nsight Systems

Computer Vision & 3D: VLMs, detection, segmentation, tracking, multi-view geometry, camera calibration, stereo reconstruction, PnP

Deployment & Infra: Core ML, TensorFlow Lite, ONNX Runtime, FastAPI, Docker, Kubernetes, AWS (EC2/S3), Linux, Git

EXPERIENCE

Pengtao Xie Lab, UC San Diego

La Jolla, CA

Student Researcher, Autonomous AI Systems – Advisor: Prof. Pengtao Xie

Apr 2026 – Present

- Building task-conditioned intervention-effect models and a cost-aware multi-fidelity experiment controller over DataMaster's UCB baseline to predict gain, cost, and failure risk of data transformations under fixed compute budgets.
- Established cross-task transfer via leave-one-task-out evaluation across 5 tasks, graph retrieval cut time-to-target **16.9%** and API cost **14.8%** versus no cross-task memory.

Player

San Diego, CA

Mobile Computer Vision Intern, Cricket Analytics

Jun 2026 – Sep 2026

- Shipped a **YOLO26** ball detector to iPhone using FP16 Core ML at **42 ms/frame** with **2%** recall loss vs. PyTorch.
- Built and annotated an **11.8K+ frame** detection benchmark from 120-fps stereo clips, improved ball-detection recall from **0.65 to 0.92** through model-scale, input-resolution, and augmentation sweeps.
- Automated **stereo camera recalibration** for a two-iPhone rig after camera movement using learned 3D scene matching and **PnP** pose estimation, eliminating manual point selection and reducing relative-rotation error from **7.95° to 0.54°**.

Medical Imaging and Reconstruction Lab, IIT Madras

Chennai, India

Research Assistant – Advisor: Prof. Ganapathy Krishnamurthi

Jun 2024 – Aug 2025

- Compressed a **DINOv2**-based Critical-View-of-Safety classifier for CPU-only deployment via LoRA fine-tuning, knowledge distillation, and **INT8 quantization**, reducing model size **344 MB→22 MB** and CPU latency **3.5×** while retaining **98.2%** of teacher accuracy.
- Adapted **SAM 2** for surgical video segmentation of hepatocystic anatomy and tools, reaching **89.4% mIoU**.
- Built a **2D-to-3D coronary reconstruction** from biplane angiography (**0.42 mm** mean error), deployed with **Meril Life Sciences**.

SELECT PROJECTS

DRISHTI – On-Device Assistive Vision System [\[Code\]](#) [\[Report\]](#)

Best Healthcare Hack, Hard Hack 2026 – Gemma 3n E2B, LiteRT-LM, TensorFlow Lite, Kotlin, Raspberry Pi

- Integrated **Gemma 3n E2B** on Android using **LiteRT-LM**, with GPU vision inference and Raspberry Pi camera input over HTTP, connected streaming output to sentence-level TTS for speech playback during generation.
- Implemented model-triggered face recognition using **ML Kit detection** and **512-D ArcFace embeddings**, matching against enrolled faces with TensorFlow Lite GPU acceleration and CPU fallback.

SurgLedger – Surgical Video Question Answering [\[Code\]](#)

MICCAI 2026 ORena FOCUS Challenge – Qwen3-VL-8B, LoRA, PyTorch DDP, vLLM, Docker

- Built a surgical video QA pipeline with **Qwen3-VL-8B**, added a second video review step that **doubled temporal localization accuracy** without additional training.
- Reduced **video decoding time by 31%** through keyframe-aligned sampling, deployed offline Docker inference at **9.4s/question**.

Attention Kernel Portability Across Six GPUs [\[Code\]](#) [\[Report\]](#)

CUDA, Triton, FlashAttention-2/3, FlashInfer, PyTorch SDPA, TorchInductor

- Built a benchmark harness for **16 attention implementations** across **6 NVIDIA GPUs**, collecting **54K+ valid timings** with correctness checks and kernel-dispatch tracing.
- Found the fastest backend changed across GPUs in **42% of prefill** versus **10% of decode** cases with clear winners (Ampere/Ada/Hopper). Reusing another GPU's pick incurred **1.61×** the target's best prefill latency at **p95**, with a median near **1.0×**.

CXRGenie – Multimodal Diagnostic Agent for Chest X-Ray Interpretation [\[Project Page\]](#) [\[Report\]](#)

B.Tech Thesis – GPT-4o-mini, LangGraph, Qdrant

- Built a **ReAct multimodal agent** orchestrating five CXR tools (segmentation, classification, grounding, VQA, report generation) over LangGraph with PubMedBERT/Qdrant retrieval, beating CheXagent and LLaVA-Med-7B on ChestAgentBench and CheXbench.
- Ablated components to isolate the gains: **70.1%** on CheXbench vs **55.2%** bare LLM, with tools worth **+9.0** points and retrieval **+6.0**, **104s** median latency traced to tool-call depth, not model inference.

PUBLICATIONS

- “Spatially Enhanced U-Net with Non-Local Attention Gates for Automated Retinal Vasculature and Foveal Avascular Zone Segmentation in OCTA Images.” Raja, N.P., **Sarvesan, S.** *Engineering Applications of Artificial Intelligence* (2026). [\[Paper\]](#)
- “Dense-PMSFNet: DenseNet Pyramidal Multi-Scale Fusion Network for Retinal Vasculature and FAZ Segmentation in OCTA.” Raja, N.P., **Sarvesan, S.**, Thomas, A., Gopi, V.P. *IEEE Latin America Transactions* (2025). [\[Paper\]](#)

AWARDS

- Best Healthcare Hack & MLH Best Use of ElevenLabs, HARD Hack, UC San Diego 2026
- IEEE AP-S Undergraduate Summer Research Scholarship – selected from 79 global applicants 2024
- Dr. Ram & Thaila Foundation Research Scholarship – institute-wide, all departments, NIT Trichy 2024
- Summer Research Fellowship, Indian Academy of Sciences – selected from 5,000+ applicants 2023